THU

Technische
Hochschule
Ulm

HENSOLDT

A Typesetting System to Untangle the Scientific Writing Process

Gabriel Zimmermann

I. Abstract

The process of scientific writing is often tangled up with the intricacies of typesetting, leading to frustration and wasted time for researchers. In this paper, we introduce Typst, a new typesetting system designed specifically for scientific writing. Typst untangles the typesetting process, allowing researchers to compose papers faster. In a series of experiments we demonstrate that Typst offers several advantages, including faster document creation, simplified syntax, and increased ease-of-use.

II. Something…

Informatik

Bachelor thesis at Ulm University of Applied Sciences
Department of Computer Science
Degree Program Informatik
Presented by Max Mustermann
April 2026

1) Reviewer: Prof. Dr.-Ing. Klaus Baer
2) Reviewer: Prof. Dr.-Ing. Philipp Graf

III. Introduction

Analysis of dynamic plug-in systems in rust. Comparison of different approaches focusing the pros and cons.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri. [1]

IV. Wawa

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat.

adadad

exactly uwu

V. Methods

Goal of the research was to analyse the following points:

- performance
- development complexities
- dependencies
- safety
- interoperability

A. Performance

Only "pure" quantitative measurement will be the performance and maybe safety (using safety levels).

a) *Tools:*

Using criterion and gungraun benchmarks. Not a perfect measurement, but it should give us enough hints about what happens behind the scenes.

Using black_box, we can ensure that the iternal code isn't hyper-optimized by the compiler, which can lead to more accurate benchmarks.

b) *Benchmarks:*

To test all of this, we use a simple temperature simulation, where an random amount of sensors will pick up the data.

Benches are run with the follolwing set of configurations:

- run app "regarulary" with black_box
- run average temperature fusion code with black_box once (sensors: 1 vs. 1000000)
- run average temperature fusion code without black_box (sensors: 1 vs. 1000000)

c) *Evaluation:*

Gunguan helps us see the internal required instructions on "my" CPU. This result should correlate with the actual run time using criterion. We can then reason about the expected overhead of a choosen approad.

B. Development complexities

Schulnote

Qualitative Meaurement

C. Limitations

Might be part of "complexities"??

Qualitative Meaurement

D.

References

[1] L. Morris and R. Astley, "Net Wok++: Taking distributed dumplings to the cloud," *Armenian Journal of Proceedings*, vol. 65, pp. 101–118, 2022.